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March 20, 2026

David Landsman, Editor-in-Chief
Database: The Journal of Biological Databases and Curation
Oxford University Press

Re: Submission of Database Resource Paper - “KinaseCanvas: An Integrated Resource for Curated and Novel Kinase-Substrate Interactions”

Dear Dr. Landsman,

We are pleased to submit our database resource paper describing **KinaseCanvas**, an open-access resource that directly addresses a critical curation gap in the phosphoproteomics community by dramatically expanding the functional annotation of the understudied “dark” kinome.

Currently, curated kinase databases are heavily skewed toward well-studied signaling hubs, while recent kinome-wide specificity atlases generate millions of raw predictions that are too noisy for direct interpretation. KinaseCanvas bridges this gap. By applying three stringent biological filters—motif rank, Human Protein Atlas co-localization, and machine learning-derived functional scores—we successfully integrate empirical motif predictions with known interactions to yield a high-confidence, context-aware network.

Key Impacts and Discoveries:

- **Large Network Expansion:** We generate **13,048 novel kinase-substrate pairs** not present in existing databases, representing a 56% expansion of coverage (428 kinases and 4,614 substrates).
- **Illuminating the Dark Kinome:** We triple the representation of understudied “dark” kinase interactions (from 3.4% to 11.2%), addressing a major priority in kinase biology.
- **Biological Utility:** KinaseCanvas enables rapid hypothesis generation. For example, our integration highlights a novel ABL1→ARID1A phosphorylation event with actionable implications for cancer biology.

Resource Accessibility and Sustainability: KinaseCanvas is FAIR-compliant and features an interactive web application (KinaseCanvas Explorer) for dynamic filtering by confidence and functional evidence. All underlying data (including >28 million raw specificity scores), code, and documentation are publicly available via GitHub (<https://cujoisa.github.io/KinaseCanvas/>). We are committed to 5+ years of automated updates and institutional maintenance.

Thank you for your time and consideration.

Sincerely,

Chinmaya Joisa
Postdoctoral Fellow
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